Chapter 5 Chemical reactions

EXTRA WORKSHEET 2: Ionic compounds and precipitation reactions

Ionic compounds form crystal lattices of cations and anions. Below in Table 1 is a list of some common cations and anions that make up ionic compounds.

Table 1

Positive ions (cations)		Negative ions (anions)	
Chemical name	**Symbol**	**Chemical name**	**Symbol**
Hydrogen ion	H^+	Fluoride	F^-
Lithium ion	Li^+	Chloride	Cl^-
Sodium ion	Na^+	Bromide	Br^-
Potassium ion	K^+	Iodide	I^-
Copper(I) ion	Cu^+	Hydroxide	OH^-
Calcium ion	Ca^{2+}	Nitrate	NO_3^-
Magnesium ion	Mg^{2+}	Oxide	O^{2-}
Barium ion	Ba^{2+}	Sulfide	S^{2-}
Copper(II) ion	Cu^{2+}	Sulfate	SO_4^{2-}
Iron(II) ion	Fe^{2+}	Carbonate	CO_3^{2-}
Iron(III) ion	Fe^{3+}	Nitride	N^{3-}
Aluminium ion	Al^{3+}	Phosphate	PO_4^{3-}

Naming ionic compounds

Naming ionic compounds is relatively simple. The naming conventions follow just two rules:

1 The chemical name is made up of the name of the cation followed by the name of the anion.
Example: an ionic compound made up of potassium cations and fluoride anions is known as potassium fluoride.

2 Ions from the same atom but with a different charge are distinguished by roman numerals that show the charge of the ion. These roman numerals are included in the chemical name.

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Example: Iron atoms form two ions – iron(II), Fe^{2+} and iron(III), Fe^{3+}. These both form oxides known as iron(II) oxide and iron(III) oxide.

Questions

1 Use the rules and the table above to state the chemical name for the following ionic compounds:

a $MgCl_2$

_____

b Na_2O

_____

c Ca_3N_2

_____

d $CuCO_3$

_____

e $Al(OH)_3$

_____

f FeO

_____

g $Ca_3(PO_4)_2$

_____

2 Write the chemical formulas for the following ionic compounds:

a lithium carbonate _____

b aluminium bromide _____

c copper(II) nitrate _____

d sodium chloride _____

e calcium iodide _____

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f ammonium hydroxide _____

g lithium nitride _____

Predicting precipitation reactions

Some ionic compounds are soluble in water (dissolve) and some ionic compounds are insoluble in water (do not dissolve). This means that when two aqueous solutions of soluble ionic compounds are mixed, the ions may exchange to form an insoluble compound. When this occurs, the insoluble compound comes out of solution as a solid.

This type of chemical reaction is known as a precipitation reaction and the insoluble solid is known as the precipitate. Scientists can predict if mixing two solutions will result in a precipitate by using solubility rules.

Some solubility rules are shown in Table 2. These rules tell you which combinations of anions and cations are soluble and which combinations are insoluble.

Table 2

Soluble	**Exceptions**	
	Insoluble	**Slightly soluble**
Most chlorides	AgCl,	$PbCl_2$
Most bromides	AgBr,	$PbBr_2$
Most iodides	AgI, PbI_2	
All nitrates	No exceptions	
Most sulphates	$SrSO_4$, $BaSO_4$, $PbSO_4$	$CaSO_4$, Ag_2SO_4

Insoluble	**Exceptions**	
	Soluble	**Slightly soluble**
Most hydroxides	NaOH, KOH, $Ba(OH)_2$	$Ca(OH)_2$, $Sr(OH)_2$
Most carbonates	Na_2CO_3, K_2CO_3, $(NH_4)_2CO_3$	
Most phosphates	Na_3PO_4, K_3PO_4, $(NH_4)_3PO_4$	

Questions

1 Which of the following compounds are soluble and which are insoluble?

a $Pb(NO_3)_2$ **g** $Ba(OH)_2$

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b	$Al(OH)_3$	**h**	$CaCO_3$
c	Na_2SO_4	**i**	AgBr
d	CaS	**j**	$Pb(NO_3)_2$
e	$PbCl_2$	**k**	$FeCl_3$
f	$FePO_4$	**l**	$SrSO_4$

3 Use the solubility rules to determine if the following reactions result in a precipitate and if so, the chemical name and formula of the precipitate.

a $Pb(NO_3)_2 + KCl \rightarrow PbCl_2 + KNO_3$ Precipitate: YES / NO

Chemical name: _____ Formula: _____

b $NaOH + Al_2(SO_4)_3 \rightarrow Na_2SO_4 + Al(OH)_3$ Precipitate: YES / NO

Chemical name: _____Formula: _____

c $AgNO_3 + Na_2S \rightarrow Ag_2S + NaNO_3$ Precipitate: YES / NO

Chemical name: _____Formula: _____

d $NaCl + KNO_3 \rightarrow NaNO_3 + KCl$ Precipitate: YES / NO

Chemical name: _____Formula: _____

e $Na_2SO_4 + BaCl_2 \rightarrow NaCl + BaSO_4$ Precipitate: YES / NO

Chemical name: _____Formula: _____

e $Li_2CO_3 + Al(NO_3)_3 \rightarrow LiNO_3 + Al_2(CO_3)_3$ Precipitate: YES / NO

Chemical name: _____Formula: _____

4 **i** Write word equations to describe the chemical reactions that could occur when the following solutions are mixed.

ii Write formula equations for each of the word equations.

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iii Determine if the reaction would result in a precipitate and if so, state the chemical name of the precipitate.

a Solutions of calcium chloride and potassium phosphate

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

b Solutions of calcium nitrate and sodium chloride

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

c Solutions of potassium chloride and silver nitrate

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

d Solutions of nickel(II) sulphate and sodium sulphide

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

e Solutions of iron(II) nitrate and potassium sulphide

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

f Solutions of potassium hydroxide and iron (III) nitrate

i_____

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ii _____

iii Precipitate: YES / NO Chemical name: _____

g Solutions of potassium sulphate and lead (II) nitrate

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

h Solutions of calcium chloride and sodium phosphate

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

i Solutions of potassium nitrate and magnesium iodide

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

j Solutions of iron(II) nitrate and barium hydroxide

i_____

ii _____

iii Precipitate: YES / NO Chemical name: _____

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